👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 3 · STANDARD 6.AT.2

Understand Rates and Unit Rates

Lesson 3-2 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

ARCADE BUILDER MISSION

Best Booth in the Arcade

You are the new operations manager at NeonPlex Arcade. Two ticket booths are competing for players — Booth A charges $3 for 5 games, Booth B charges $5 for 8 games. Players want the best deal, and your job is to crown the winning booth by mastering unit rates.

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can describe a rate using its two different units and use a model to find a travel time, and I can find a unit rate to compare prices and decide the better buy.

Language Objective: I can explain a rate like "150 miles per hour" using distance and time units, and explain my choice using the words rate, unit rate, per, and better buy.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Rates and Unit Rates | **Rate** | **Unit Rate** | **Per** | **Ratio** | **Unit price** | **Better Buy** | **Divide** | **Constant** | **Total price**

Level 3: try the blanks from memory first, then check the bank.

1. A rate compares different units; a _____ _____ compares a quantity to exactly 1 unit.

2. A ratio that compares two quantities with different units is a _____ .

3. A rate that tells the cost or amount for one unit is a _____ .

4. The word that means 'for each one' is _____ .

5. A comparison of two quantities is a _____ .

6. A unit rate that tells the cost of one item is the _____ _____ .

7. The option with the lower unit rate, costing less per item, is the _____ .

8. To find a unit rate, I _____ the price by the number of units.

9. A rate that never changes is _____ .

10 The cost of the whole package is the _____ .

Write About the Math

Which size is the better deal, and how do you know?
¿Qué tamaño es la mejor oferta y cómo lo sabes?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Rates and Unit Rates** — *Tasas y tasas unitarias*
- ☐ **Rate** — *Tasa*
- ☐ **Unit Rate** — *Tasa unitaria*
- ☐ **Per** — *Por*
- ☐ **Ratio** — *Razón*

Model: *It is hard to compare because the booths have different numbers of games.* — Students notice the prices cannot be compared directly because the number of games differs, so they need the cost for one game (the unit rate) to compare fairly.

Start

Complete the frames. Use the word bank.
Completa las oraciones. Usa el banco de palabras.

- **The ___ bag is the better deal because its unit rate is $___ per cup, which is ___ than $___ per cup for the other size.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

_____

_____

_____

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.

- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Rates and Unit Rates
2. **Notes 2:** Rate
3. **Notes 3:** Unit Rate
4. **Notes 4:** Per
5. **Notes 5:** Ratio
6. **Notes 6:** Unit price
7. **Notes 7:** Better Buy
8. **Notes 8:** Divide
9. **Notes 9:** Constant
10. **Notes 10:** Total price
11. **Watch for this mistake:** *A common mistake in Rates and Unit Rates is comparing total costs instead of unit rates: a student sees Pack X at $4.00 for 10 tokens and Pack Y at $6.00 for 16 tokens and picks Pack X just because $4.00 is the smaller number, without dividing to find the cost per token ($0.40 for Pack X vs. $0.375 for Pack Y — Pack Y is actually the better buy). A second common mistake is dividing backwards — finding tokens ÷ cost instead of cost ÷ tokens — which turns 'dollars per token' into 'tokens per dollar' and gives an answer that looks reasonable but answers the wrong question. Always divide the total cost by the number of items, and check that your unit rate answers 'the cost for ONE item,' not the reverse.*
12. **Try It 1:** A. Pulse, at 15 points per minute — *Nova: 78 ÷ 6 = 13 points per minute. Pulse: 75 ÷ 5 = 15 points per minute. 15 > 13, so Pulse is faster.*
13. **Try It 2:** B. Machine B, at 21 coins per play — *Machine A: 150 ÷ 5 = 30 coins per play. Machine B: 168 ÷ 8 = 21 coins per play. 21 < 30, so Machine B is the better value.*